



What Is a Bug?

Ontario Math C3.2 — read, find, and fix code

Name: _____

Date: _____

★ I can find a bug in a loop and fix the repeat number.

A bug is a mistake in code

A bug is a mistake that makes code do the wrong thing. Debugging is finding and fixing it.

Use 3 steps: say what the code **SHOULD** do, see what it **DOES**, then **FIX** the wrong part. Today the bug is the repeat number.

How to debug



Debug in 3 steps: SHOULD → DOES → FIX.

Bit should JUMP 5 times — but this is buggy

when green flag clicked

repeat 3

jump

It should be 5 jumps, but repeat 3 makes only 3. Fix the repeat number to 5.

1 Find the bug

Bit should CLAP 6 times. The code says repeat 4. What is wrong, and what should the repeat number be? _____

2 Too many this time

Bit should MOVE 4 times, but the code says repeat 7. Is that too many or too few? What should it be?

3 Write the fix

Bit should HOP 8 times. The code says repeat 2. Write the repeat number that fixes it: repeat _____

4 Challenge

Bit should clap 6 times. The loop has TWO claps inside and says repeat 6 — so Bit claps 12! What repeat number gives exactly 6 claps? Explain.

Big idea

To debug: say what the code SHOULD do, compare it to what it DOES, and change the wrong part. A wrong repeat number is the most common loop bug.